

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

We discuss about the possibilities of a cyborg or robot takeover in the future and we also delve into understanding Earth's newest pollutant, plastitar and how it affects the planet.



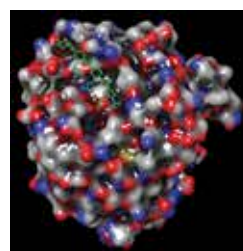
A dynamic duo

Scientists at the University of Illinois Chicago have successfully identified two new lung blood vessel cells. Of the two, one has been found to express more genes involved in inflammation and regulating immune response, while the other expresses more genes involved in cell regeneration and reproduction. <https://dgit.in/DynamicDuo>

WHAT'S NEW

Thank goodness!

A new molecule, capable of killing, hard-to-treat cancer cells has been successfully synthesised, bringing in a new ray of hope for cancer patients. Dr Jung-Mo Ahn, an associate professor of Chemistry and Biochemistry at UT Dallas, is the researcher behind this ground-breaking discovery. His study, titled "Targeting LIPA independent of its lipase activity is a therapeutic strategy in solid tumours via induction of endoplasmic reticulum stress", containing all the information about this new molecule, was published in the journal Nature Cancer. The molecule is called ERC-41 and can kill an array of hard-to-treat cancer cells. This included triple-negative breast cancer, pancreatic and ovarian cancers, and glioblastoma cells.



This newly synthesised chemical achieves this by "exploiting a weakness in cells not previously targeted by other drugs". During testing, this molecule was successful

in killing cancer cells in actual human tissues. The best thing was that during the tests, no adverse effects were observed after its use.

Dr Ahn said, "The ERC-41 compound did not kill healthy cells, but it wiped out tumour cells regardless of whether the cancer cells had estrogen receptors." <https://dgit.in/CancerCure>

More stress, less immunity

A research study titled "Social stressors associated with age-related T lymphocyte percentages in older US adults: Evidence from the US Health and Retirement Study" has found that stress is responsible for speeding up the ageing of the human immune system. The study also noted that faster ageing of the immune system increased the risks of cardiovascular ailments, cancer, and illness after contracting COVID-

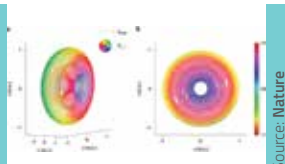


19. To understand and measure the exposure of adults to social stress, the researchers analysed responses given by a sample of 5,744 adults, all of whom were above 50 years of age. Analysis

of the blood samples of the participants was also done using a technique called cytometry. As noted in the findings, individuals with higher stress had "older-seeming immune profiles" and a relatively higher number of worn-out white blood cells.

A better diet and involvement in exercise could potentially alleviate stress, thereby reducing the ageing of the immune system.

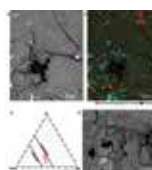
<https://dgit.in/NoMoStress>



Source: Nature

Smoking light rings

Researchers from the University of Shanghai have been able to create vortex rings using light. Having replicated a fluid-like flow of energy, the researchers now look forward to replicating the results with an electric current or a magnetic field. <https://dgit.in/ItsSmokin>



Source: Science.org

Maximum purity

Japanese researchers have found that the grains of the asteroid, Ryugu, which had been evolving for millions of years, are the "least contaminated asteroid material" on the planet. The recovered sample was also found to be mainly composed of hydrous phyllosilicate materials. <https://dgit.in/PureRock>



Silent ones are harmful

A research study focusing on the genetic manipulation of yeast cells has shown that most of the synonymous genetic mutations are harmful. Also known as 'silent' mutations, these mutations were considered mostly neutral until now. <https://dgit.in/Beware>